



NORTH SYDNEY BOYS HIGH SCHOOL

2018 HSC ASSESSMENT TASK 4

Mathematics

Extension 1

General Instructions

- Working time – 60 minutes (+ 5 minutes reading time)
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Dr Jomaa
- ☐ Ms Lee
- ☐ Ms Ziazaris

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	Total	Total %
Mark	$\overline{12}$	$\overline{9}$	$\overline{10}$	$\overline{6}$	$\overline{37}$	$\overline{100}$

Question 1 – Binomial Theorem (12 Marks)

- (a) Expand and simplify:

$$(3x - 2y)^5 \quad 2$$

- (b) Find the coefficient of x^3 in.

$$\left(x^2 + \frac{3}{x}\right)^9 \quad 2$$

- (c) Find the term independent of x . Express your answer fully simplified in integer form.

$$(3 - x)^4 \left(1 + \frac{2}{x}\right)^7 \quad 3$$

- (d) Find the value of n such that the coefficient of x^4 is twice the coefficient of x^3 in the expansion of $(1 + x)^n$ 2

- (e) In the expansion of $(1 + x)(a - bx)^{12}$ it is known that the coefficient of x^8 is 0.

Find $\frac{a}{b}$ in simplest form. 3

Question 2 – Projectile Motion 1 (9 Marks)

- (a) A golf ball is hit at angle α , where $0^\circ < \alpha < 90^\circ$. The initial velocity of the ball is $V \text{ ms}^{-1}$. Gravity is g and time is t .

Show that the horizontal and vertical displacement of the ball is given by:

$$x = Vt \cos \alpha$$

$$y = -\frac{1}{2}gt^2 + Vt \sin \alpha$$

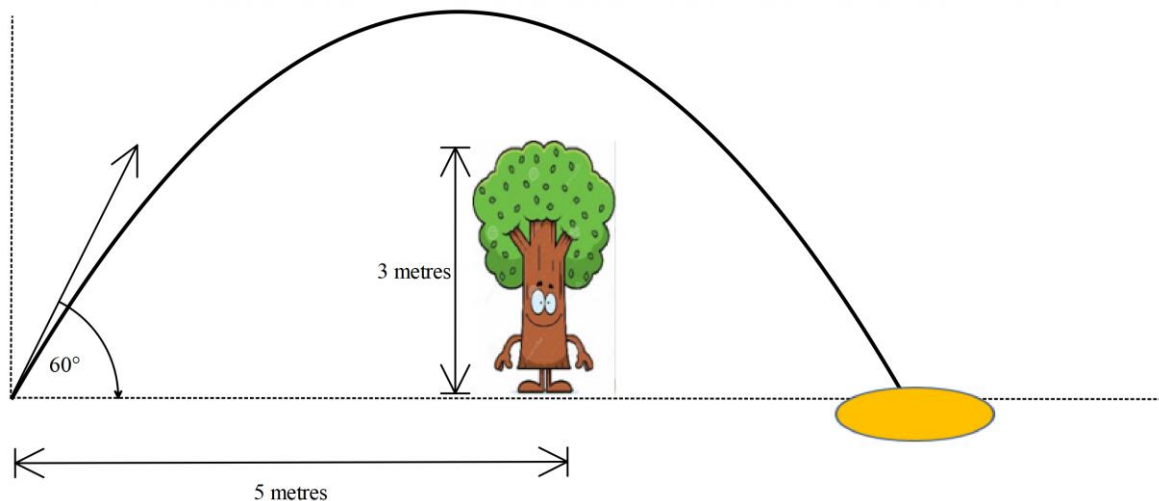
2

- (b) Show that

$$y = x \tan \alpha - \frac{gx^2}{2V^2} (1 + \tan^2 \alpha)$$

2

Mark hits his ball so that it has an initial velocity of 12 ms^{-1} and an angle of projection of 60° . Acceleration due to gravity is $g = 9.8 \text{ ms}^{-2}$.



To succeed, Mark must clear a 3m shrub which is 5m away from his ball.

- (c) How high over the shrub does the ball clear? (*i.e. how much higher than the top of the shrub is the ball when it passes directly over the shrub*)
- (d) What is the horizontal distance travelled by the ball? (*assume the ball lands in a bunker of sand and stops immediately*)

2

3

Question 3 – Simple Harmonic Motion (10 Marks)

- (a) A particle moving in simple harmonic motion about the origin has a period of $\frac{\pi}{3}$ seconds and a maximum velocity of 2 metres/second. Using the formula $v^2 = n^2(a^2 - x^2)$, or otherwise, find:
- (i) The amplitude of motion **2**
- (ii) The displacement after $\frac{\pi}{6}$ seconds (*assume the particle is initially at the origin*). **1**
- (b) The displacement x metres of a particle moving in simple harmonic motion is given by $x = 4 \sin \pi t$ where t is the time expressed in seconds.
- (i) Find the period of oscillation. **1**
- (ii) Find the speed at the centre of motion. **2**
- (iii) Show the acceleration is proportional to the displacement from the centre of motion **1**
- (iv) What is the velocity when $x = 3$? **3**

Question 4 – Projectile Motion (6 Marks)

A projectile is fired from a point on horizontal ground with initial speed $V \text{ ms}^{-1}$ and an angle of projection θ . The acceleration due to gravity is g .

Let x and y be the horizontal and vertical displacements from O , the point of projection. They are given by the following equations: **(you do not need to prove these).**

$$x = Vt \cos \theta$$

$$y = -\frac{1}{2}gt^2 + Vt \sin \theta$$

The cartesian equation of the path is: **(you do not need to prove this).**

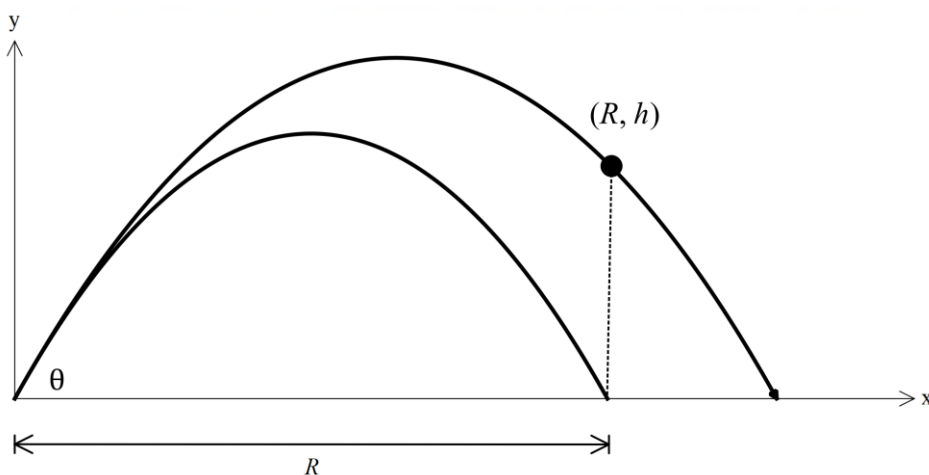
$$y = x \tan \theta - \frac{gx^2}{2V^2 \cos^2 \theta}$$

Show that the maximum range R on the horizontal plane is given by:

(a)

$$R = \frac{V^2}{g}$$

2



(b) Show that to hit a target h metres above the ground at the same horizontal distance R using the

same angle of projection θ , the speed of projection must be increased to $\frac{V^2}{\sqrt{V^2 - gh}}$.

4

Question 1

(a) $(3x - 2y)^5$

$$= \binom{5}{0}(3x)^5 - \binom{5}{1}(3x)^4(2y) + \binom{5}{2}(3x)^3(2y)^2 - \binom{5}{3}(3x)^2(2y)^3 + \binom{5}{4}(3x)(2y)^4 - \binom{5}{5}(2y)^5 \quad \checkmark$$

$$= 243x^5 - 810x^4y + 1080x^3y^2 - 720x^2y^3 + 240xy^4 - 32y^5 \quad \checkmark$$

(b) A typical term in the expansion is:

$$\binom{9}{k}(x^2)^{9-k}(3x^{-1})^k$$



$$= \binom{9}{k} 3^k x^{18-2k} x^{-k}$$

$$= \binom{9}{k} 3^k x^{18-3k}$$

$$18 - 3k = 3$$

$$3k = 15$$

$$k = 5$$

$$\therefore \binom{9}{5} 3^5 \text{ or } 30618$$



(c) A typical term in the expansion is:

$$\begin{aligned} & \binom{4}{k} 3^{4-k} (-x)^k \binom{7}{r} (1)^{7-r} (2x^{-1})^r \\ &= \binom{4}{k} \binom{7}{r} (3^{4-k}) (-1)^k 2^r x^k (x^{-1})^r \\ &= \binom{4}{k} \binom{7}{r} 3^{4-k} (-1)^k 2^r x^{k-r} \end{aligned}$$



$$k - r = 0$$

$$\therefore k = r$$



$$\binom{4}{0} \binom{7}{0} 3^4 (-1)^0 2^0 = 81$$

$$\binom{4}{1} \binom{7}{1} 3^3 (-1)^1 2^1 = -1512$$

$$\binom{4}{2} \binom{7}{2} 3^2 (-1)^2 2^2 = 4536$$

$$\binom{4}{3} \binom{7}{3} 3^1 (-1)^3 2^3 = -3360$$

$$\binom{4}{4} \binom{7}{4} 3^0 (-1)^4 2^4 = 560$$

$$305$$



$$(d) \quad 2 \binom{n}{3} = \binom{n}{4}$$

$$\frac{2 \cancel{n!}}{3! (n-3)!} = \frac{\cancel{n!}}{4! (n-4)!}$$



$$\frac{2}{\cancel{3!} (n-3) \cancel{(n-4)!}} = \frac{1}{4 \times \cancel{3!} (n-4)!}$$

$$\frac{2}{n-3} = \frac{1}{4}$$

$$\therefore n-3 = 8$$

$$n = 11$$



$$(e) \quad 1(a-bx)^{12} + x(a-bx)^{12}$$

x^8 is:

$$\binom{12}{8} a^4 (-b)^8 + \binom{12}{7} a^5 (-b)^7 = 0$$



$$\therefore \binom{12}{8} a^4 b^8 = \binom{12}{7} a^5 b^7$$

$$\therefore \frac{a^5 b^7}{a^4 b^8} = \frac{\binom{12}{8}}{\binom{12}{7}}$$



$$\therefore \frac{a}{b} = \frac{5}{8}$$



Question 2

(a) $\ddot{x} = 0$

$$\dot{x} = C$$

when $t = 0$

$$\dot{x} = V \cos \alpha$$

$$x = Vt \cos \alpha + C$$

when $t = 0$ $x = 0$

$$0 = V(0) \cos \alpha + C$$

$$\therefore C = 0$$

$$\therefore x = Vt \cos \alpha \quad \boxed{\checkmark}$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + C$$

when $t = 0$ $\dot{y} = V \sin \alpha$

$$\therefore V \sin \alpha = -g(0) + C$$

$$\therefore C = V \sin \alpha$$

$$\dot{y} = -gt + V \sin \alpha$$

$$y = -\frac{1}{2}gt^2 + Vt \sin \alpha + C$$

when $t = 0$ $y = 0$

$$0 = -\frac{1}{2}g(0)^2 + V(0) \sin \alpha + C$$

$$\therefore C = 0$$

$$\therefore y = -\frac{1}{2}gt^2 + Vt \sin \alpha \quad \boxed{\checkmark}$$

(b) $x = Vt \cos \alpha$

$$t = \frac{x}{V \cos \alpha}$$

$$y = -\frac{1}{2}g \left(\frac{x}{V \cos \alpha} \right)^2 + V \left(\frac{x}{V \cos \alpha} \right) \sin \alpha \quad \boxed{\checkmark}$$

$$= -\frac{1}{2}g \left(\frac{x^2}{V^2 \cos^2 \alpha} \right) + x \frac{\sin \alpha}{\cos \alpha}$$

$$= -\frac{1}{2}g \frac{x^2}{V^2} \sec^2 \alpha + x \tan \alpha$$

$$= -\frac{g x^2}{2V^2} (1 + \tan^2 \alpha) + x \tan \alpha$$

$$= x \tan \alpha - \frac{g x^2}{2V^2} (1 + \tan^2 \alpha) \quad \boxed{\checkmark}$$

(c) when $x = 5$, $\alpha = 60^\circ$, $v = 12$ $g = 9.8$

$$y = 5 \tan 60^\circ - \frac{9.8 \times 5^2}{2 \times 12^2} (1 + \tan^2 60^\circ) \quad \boxed{\checkmark}$$

$$= 5\sqrt{3} - \frac{245}{288} (4)$$

$$= 5.257476 \dots$$

$$\therefore \text{clearance} = 5.257 \dots - 3$$

$$= 2.26 \text{ m (3 s.f.)}$$



$$(d) y = x \left(\tan \alpha - \frac{gx}{2v^2} (1 + \tan^2 \alpha) \right)$$

when $y = 0$

$$x = 0 \quad \text{or} \quad \tan \alpha - \frac{gx}{2v^2} (1 + \tan^2 \alpha) = 0$$



as $x \neq 0$ at max distance

$$\tan \alpha - \frac{gx}{2v^2} (1 + \tan^2 \alpha) = 0$$

$$\therefore \frac{gx}{2v^2} (1 + \tan^2 \alpha) = \tan \alpha$$

$$\therefore x = \frac{2v^2 \tan \alpha}{g(1 + \tan^2 \alpha)}$$



$$= \frac{2 \times 12^2 \times \sqrt{3}}{9.8(1 + 3)}$$

$$= 12.7 \text{ m (3 s.f.)}$$



Question 3

$$(a) (i) \frac{2\pi}{n} = \frac{\pi}{3}$$

$$\therefore n = 6 \quad \boxed{\checkmark}$$

$$v^2 = n^2(a^2 - x^2)$$

$$\text{when } x = 0, v = 2$$

$$2^2 = 6^2(a^2 - 0^2)$$

$$\therefore a^2 = \frac{2^2}{6^2}$$

$$\therefore a = \pm \frac{1}{3}$$

$$\therefore \text{amplitude} = \frac{1}{3} \quad \boxed{\checkmark}$$

$$(ii) \text{ Displacement is } 0 \quad \boxed{\checkmark}$$

$\left(\frac{\pi}{6}\right)$ is half a period so it
will be back at the origin)

$$(b)(i) \frac{2\pi}{\pi} = 2 \text{ seconds} \quad \boxed{\checkmark}$$

$$(ii) x = 4 \sin \pi t$$

$$\dot{x} = 4\pi \cos \pi t \quad \boxed{\checkmark}$$

at centre of motion $\cos \pi t = 1$

$$\therefore \text{speed} = 4\pi \text{ m/s} \quad \boxed{\checkmark}$$

$$(iii) \ddot{x} = -4\pi^2 \sin \pi t$$

$$= -\pi^2 4 \sin \pi t$$

$$= -\pi^2 x \quad \boxed{\checkmark}$$

$$\therefore \ddot{x} \propto x$$

$$(iv) v^2 = n^2 (a^2 - x^2)$$

$$x = 3, \quad a = 4, \quad n = \pi \quad \boxed{\checkmark}$$

$$v^2 = \pi^2 (4^2 - 3^2)$$

$$= \pi^2 \times 7$$

$$\therefore v = \pm \pi \sqrt{7} \text{ m/s} \quad \boxed{\checkmark} \quad \boxed{\checkmark}$$

Question 4

(a) Max range occurs when $y = 0$.

$$\therefore x \tan \theta - \frac{g x^2}{2 v^2 \cos^2 \theta} = 0$$

$$\therefore x \left(\tan \theta - \frac{g x}{2 v^2 \cos^2 \theta} \right) = 0$$

as $x \neq 0$ (assuming $\theta \neq 90^\circ$)

$$\tan \theta - \frac{g x}{2 v^2 \cos^2 \theta} = 0.$$



$$\therefore \frac{g x}{2 v^2 \cos^2 \theta} = \tan \theta$$

$$\therefore x = \frac{2 v^2 \cos^2 \theta \tan \theta}{g}$$

$$x = \frac{2 v^2 \cancel{\cos^2 \theta} \frac{\sin \theta}{\cancel{\cos \theta}}}{g}$$

$$x = \frac{2 v^2 \sin \theta \cos \theta}{g}$$

$$x = \frac{v^2 \sin 2\theta}{g}$$

as the max value for $\sin 2\theta$ is the max range is.

$$R = \frac{v^2}{g}$$



(b) let the new velocity be w .

Sub $y = h$, $x = R$, $\theta = 45^\circ$ & $V = w$

$$\text{into } y = x \tan \theta - \frac{g x^2}{2 v^2 \cos^2 \theta}$$



$$\therefore h = R \tan 45^\circ - \frac{g R^2}{2 w^2 (\cos 45^\circ)^2}$$

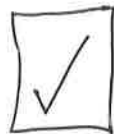
$$h = R(1) - \frac{g R^2}{\cancel{2} w^2 (\cancel{\frac{1}{\sqrt{2}}})^2}$$

$$h = R - \frac{g R^2}{w^2}$$

$$\therefore \frac{g R^2}{w^2} = R - h$$



$$\therefore w^2 = \frac{g R^2}{R - h}$$



$$\text{but } R = \frac{v^2}{g}$$

$$\therefore w^2 = \frac{g \left(\frac{v^2}{g} \right)^2}{\frac{v^2}{g} - h}$$

$$= \frac{\left(\frac{v^4 \cancel{g}}{g^2} \right)}{\left(\frac{v^2 - g h}{g} \right)}$$

$$= \frac{\left(\frac{V^4}{g}\right)}{\left(\frac{V^2 - gh}{g}\right)}$$

$$= \frac{V^4}{V^2 - gh}$$

$$\therefore \omega = \pm \sqrt{\frac{V^4}{V^2 - gh}}$$

$$= \pm \frac{V^2}{\sqrt{V^2 - gh}}$$

but as $\omega > 0$,

$$\omega = \frac{V^2}{\sqrt{V^2 - gh}}$$

